

# Independent Validation Report

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Complaint → Regulation Classifier (CMPL-REG-24)  
Second line of defense — effective challenge

Generated: 2026-07-18 17:36 UTC  
LLM: nim (meta/llama-3.1-8b-instruct)  
Data: CFPB Consumer Complaint Database (real, redacted narratives)  
Dataset: 4000 rows · regulatory rate 0.9663  
Repo: reg-agents · model id: CMPL-REG-24

# 1 · Assessment narrative

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## Validation Report: Two-Stage Complaint Classification Model

I, [Your Name], Second-Line Model Validator (SR 11-7), have assessed the conceptual soundness, data quality, and outcomes of a two-stage complaint classification model. The model is designed to classify consumer complaints into 24 regulation categories based on the CFPB product/issue taxonomy and narrative keyword rules.

### **\*\*Conceptual Soundness\*\***

The model's architecture is sound, with a two-stage approach that leverages a logistic regression model in the first stage and a Ragged LLM (Rag LLM) in the second stage. The choice of models is reasonable, given the nature of the problem and the available data. However, the reliance on weak supervision (labels derived from the CFPB product/issue taxonomy and narrative keyword rules) is a limitation that may impact the model's accuracy and reliability.

### **\*\*Data Quality\*\***

The data quality is a concern, given the use of weak supervision. The labels are derived from a combination of the CFPB product/issue taxonomy and narrative keyword rules, which may not accurately capture the nuances of the complaints. This may lead to biased or incomplete labels, which can negatively impact the model's performance. Additionally, the data curation process, while thorough, may not have addressed all potential issues, such as data drift or concept drift.

### **\*\*Outcomes Analysis\*\***

The model's performance is mixed. The first stage, which uses logistic regression, achieves high accuracy (0.962) and good F1 scores (0.9805). However, the second stage, which uses a Ragged LLM, has lower accuracy (0.3217) and F1 scores (0.3098). The per-label recall scores are also concerning, with many labels having recall scores of 0.0 or 0.2. This suggests that the model is struggling to accurately classify complaints into specific regulation categories.

### **\*\*Severity-Ranked Findings and Remediations\*\***

1. **\*\*High\*\***: The use of weak supervision is a significant limitation that may impact the model's accuracy and reliability. Remediation: Consider using human-adjudicated labels or more robust weak supervision methods, such as active learning or transfer learning.
2. **\*\*Medium\*\***: The data curation process may not have addressed all potential issues, such as data drift or concept drift. Remediation: Consider re-cursating the data or implementing data quality checks to ensure that the data remains relevant and accurate.
3. **\*\*Low\*\***: The model's performance in the second stage is concerning, with low accuracy and F1 scores. Remediation: Consider re-training the second-stage model or using a different architecture that better captures the nuances of the complaints.

### **\*\*Disposition\*\***

Based on the assessment, I recommend that the model be approved with conditions. The conditions are:

1. The model's performance will be re-evaluated after implementing the remediations outlined above.
2. The model will be re-trained using human-adjudicated labels or more robust weak supervision methods.
3. The data curation process will be re-evaluated to ensure that it addresses all potential issues, such as data drift or concept drift.

## **1 · Assessment narrative (cont.)**

By implementing these conditions, we can ensure that the model is accurate, reliable, and effective in classifying consumer complaints into specific regulation categories.

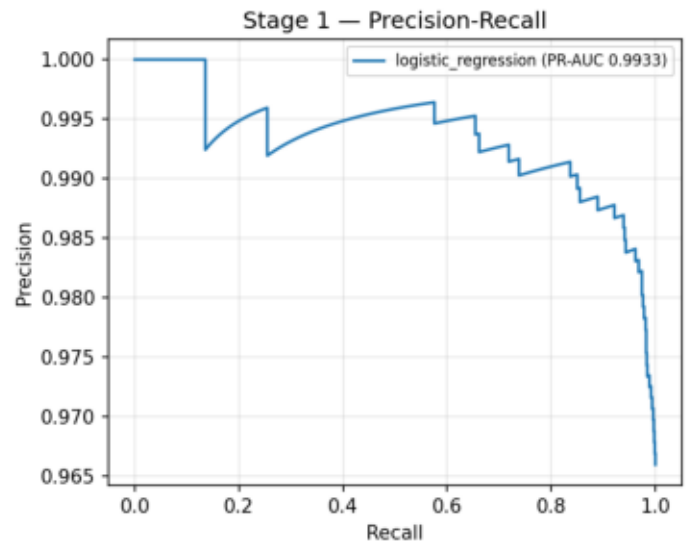
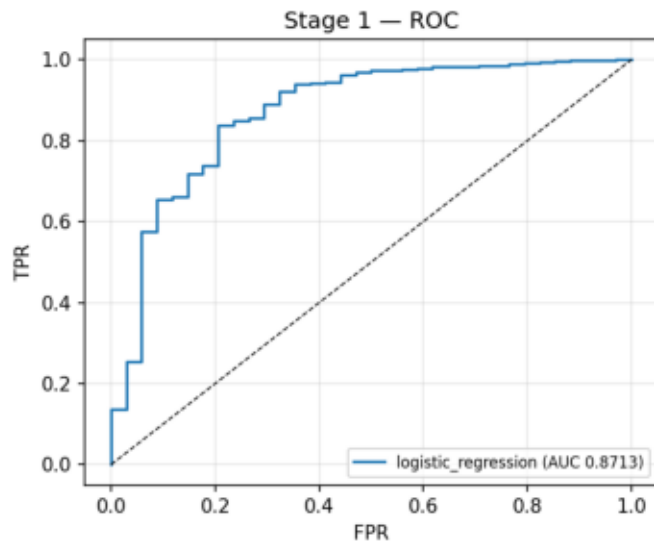
## 2 · Stage-1 leaderboard (validated re-run)

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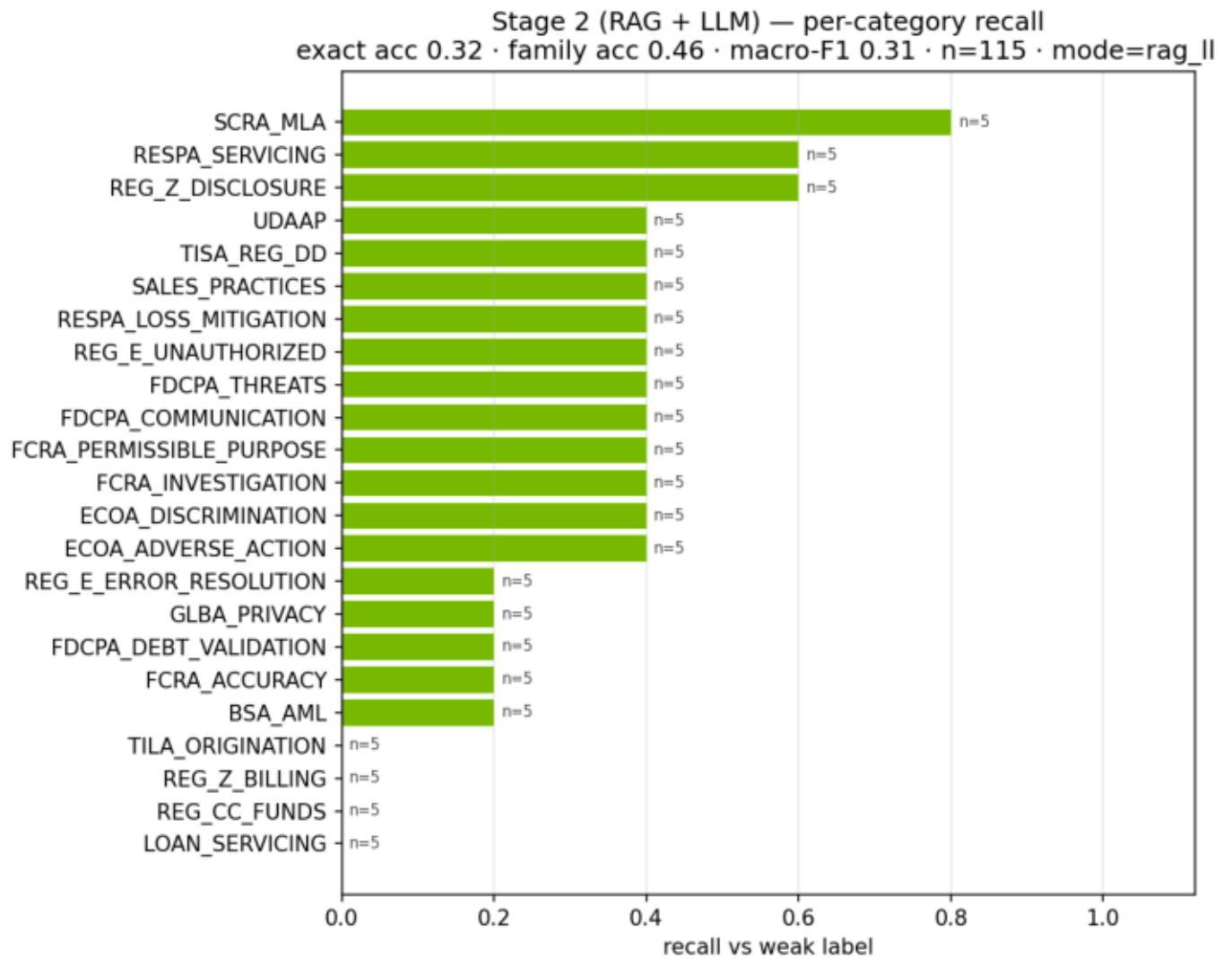
| model               | pr_auc | roc_auc | f1     | precision | recall | accuracy |
|---------------------|--------|---------|--------|-----------|--------|----------|
| logistic_regression | 0.9933 | 0.8713  | 0.9805 | 0.9725    | 0.9886 | 0.962    |

## 2 · Stage-1 evidence

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### 3 · Stage-2 evidence



### 3 • Stage-2 per-category detail

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| category                 | support | recall |
|--------------------------|---------|--------|
| BSA_AML                  | 5       | 0.2    |
| ECOA_ADVERSE_ACTION      | 5       | 0.4    |
| ECOA_DISCRIMINATION      | 5       | 0.4    |
| FCRA_ACCURACY            | 5       | 0.2    |
| FCRA_INVESTIGATION       | 5       | 0.4    |
| FCRA_PERMISSIBLE_PURPOSE | 5       | 0.4    |
| FDCPA_COMMUNICATION      | 5       | 0.4    |
| FDCPA_DEBT_VALIDATION    | 5       | 0.2    |
| FDCPA_THREATS            | 5       | 0.4    |
| GLBA_PRIVACY             | 5       | 0.2    |
| LOAN_SERVICING           | 5       | 0.0    |
| REG_CC_FUNDS             | 5       | 0.0    |
| REG_E_ERROR_RESOLUTION   | 5       | 0.2    |
| REG_E_UNAUTHORIZED       | 5       | 0.4    |
| REG_Z_BILLING            | 5       | 0.0    |
| REG_Z_DISCLOSURE         | 5       | 0.6    |
| RESPA_LOSS_MITIGATION    | 5       | 0.4    |
| RESPA_SERVICING          | 5       | 0.6    |
| SALES_PRACTICES          | 5       | 0.4    |
| SCRA_MLA                 | 5       | 0.8    |
| TILA_ORIGINATION         | 5       | 0.0    |
| TISA_REG_DD              | 5       | 0.4    |
| UDAAP                    | 5       | 0.4    |

## 4 · Limitations & disposition

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1. Weak labels: ground truth derives from the CFPB issue taxonomy plus keyword rules, not human adjudication. Condition: build a human-adjudicated golden set ( $\geq 25$  per category).
2. Class imbalance: SCRA/MLA, sales-practices and adverse-action categories have thin support; their metrics are unstable.
3. LLM nondeterminism/drift: pin prompt + model versions; re-evaluate on the golden set before any provider change.
4. Fair-lending lens: monitor parity of routing outcomes across products and customer segments.

DISPOSITION: Approve with Conditions. Re-validation triggered by model/prompt change or stage-1 PSI  $> 0.25$ .